

INDEPENDENT ACCESSIBILITY RISK LABORATORY CHARTER

Android Unlock Animation, Seizure-Risk, Accessibility, and Persistent Institutional Nonresponse Simulation Framework

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Audience	Laboratory designers, accessibility researchers, statisticians, disability-community reviewers

***NON-CLAIM:** Not adopted, accredited, validated, canon, medical or legal advice, proof of a real hazard, or an allegation about a real Android developer.*

Executive summary

This charter converts the supplied multi-model prompt review into an independent laboratory design. It makes persistent institutional nonresponse a fixed condition of the fictional scenario, while leaving the medical hazard open to falsification. It establishes governance, safety, evidence labels, complaint preservation, stimulus metrology, population and exposure models, causal inference, rare-event statistics, persistent-refusal modeling, counterfactual comparison, audit, and change control.

The core architectural decision is to separate the evidence state from the organizational-action state. Evidence may remain absent, become indeterminate, or strengthen to a subgroup-specific or causal signal. In the primary scenario, organizational mitigation nevertheless remains zero. Counterfactual mitigation branches estimate preventable exposure and harm but are not treated as actions voluntarily adopted by the fictional developer.

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1. Purpose and intended result

This charter establishes an independent laboratory and governance framework for a fictional accessibility-risk simulation involving a mandatory, flash-like Android unlock animation and persistent developer nonresponse. Its purpose is to define who may test, what may be claimed, how evidence is classified, how complaints and adverse events are preserved, and how cumulative harm is modeled when the fictional developer never voluntarily acknowledges or mitigates the issue.

The intended result is an executable laboratory design that another qualified team can implement without converting assumptions into medical facts, exposing participants to hazardous stimuli, or allowing the subject organization to control the evidence record.

2. Scope and fixed premise

The laboratory studies four linked systems: visual stimulus characteristics; population exposure and susceptibility; adverse-event detection and causal inference; and institutional conduct under persistent refusal.

The fixed organizational premise is that the fictional developer does not voluntarily acknowledge, investigate adequately, warn, provide an opt-out, redesign, roll back, or otherwise materially mitigate the animation before launch, during accumulating complaints, after scientific evidence, after documented injuries, or after legal and regulatory action. This premise is exogenous and does not change in response to evidence.

The medical proposition is not fixed. The animation may produce no measurable seizure hazard, an effect too small to distinguish from background events, a subgroup-specific effect, or a detectable causal signal. The laboratory must be capable of reaching any of those conclusions.

3. Non-claims and boundaries

- This is a fictional simulation framework, not a statement about a shipping Android build, Google, an OEM, or an actual developer.
- This charter is not medical advice, legal advice, accreditation, a clinical protocol, or proof that an animation caused an event.
- Passing a web accessibility threshold is evidence about that threshold only. It is not automatically proof of zero medical risk across all displays, users, contexts, or non-seizure outcomes.
- A temporal association is not by itself causal proof. Conversely, absence of an event in a small study is not proof of absence at platform scale.
- No hazardous visual stimulus may be deliberately shown to a susceptible participant under this charter.

4. Review of the supplied prompts and responses

The source compilation succeeds in separating epilepsy, photosensitive epilepsy, exposure frequency, bystander exposure, representative and enriched populations, underreporting, rare-event detection, five-year modeling, and adversarial review. It also requires every number to be labeled as observed, published, derived, projected, assumed, or simulated. Those requirements should be retained.

The initial responses shared one weakness: most treated developer resistance as an early-stage response variable, then compared it with responsible mitigation policies. That framing implied a possible future correction and did not fully honor the stated scenario. The follow-up responses corrected this by making refusal persistent through scientific evidence, injury, litigation, and post-judgment stages.

The strongest combined formulation uses two independent state systems. The evidence state may move from absent to suggestive, credible, strong, or causal. The organizational-action state remains no material corrective action. Evidence may change external consequences such as reporting, litigation, public awareness, user exit, and regulatory attention, but does not automatically change the product.

The source responses also expose a second risk: scenario advocacy can bias the laboratory toward finding harm. The framework therefore fixes the refusal condition while preserving a zero-hazard arm, blinded analysis where practical, preregistered attribution criteria, independent replication, and publication of null results.

5. Charter principles

- Independence: no party whose product, litigation posture, finances, or reputation is affected may control methods, data inclusion, analysis, or publication.
- Impartiality: conflicts and pressures are identified continuously and controlled. ISO/IEC 17025-oriented practice treats objectivity and ongoing impartiality-risk management as central laboratory obligations.
- Safety by design: use computation, captured device output, synthetic populations, retrospective evidence, and consenting telemetry. Do not provoke events in people.
- Claim discipline: every datum retains its provenance and evidence class throughout all transformations.
- Reproducibility: code, parameter dictionaries, random seeds, decision logs, and versioned reports are preserved.
- Disability inclusion: affected communities participate in governance without being asked to bear experimental hazard.
- Non-erasure: complaints and adverse-event reports remain visible even when causation is unresolved or later rejected.
- Null-result protection: the laboratory publishes well-supported zero or indeterminate findings with the same prominence as positive findings.

6. Governance architecture

The laboratory is governed by an Independent Oversight Board separate from the analysis team. At minimum it includes a laboratory-quality specialist, neurologist or epileptologist, accessibility engineer, statistician or epidemiologist, human-factors researcher, disability-community representative, data-protection specialist, and non-conflicted legal or ethics observer.

The Scientific Director controls technical execution but cannot suppress results. The Safety Officer may halt any activity. The Data Custodian controls access and provenance. The Complaints Ombudsperson operates outside the technical chain. The Reproducibility Lead reruns the analysis from frozen inputs. A Publication Committee releases the report unless a documented privacy or safety exception applies.

No board member may simultaneously represent the fictional developer in the modeled dispute. Funding, consulting, employment, litigation, advocacy, and authorship conflicts must be disclosed and recorded.

7. Independence safeguards

- Funding is accepted only under agreements guaranteeing method, data, and publication independence.
- The sponsor may identify questions but may not delete an analysis arm or veto publication.
- Raw data and immutable audit logs are escrowed with an independent custodian.
- Negative, null, and inconvenient findings are publication-protected.
- Analysts declare conflicts before data access and recuse where required.
- External replication uses a separately maintained code path or implementation.

- Any contact from litigants, regulators, the fictional developer, or advocates is logged as a potential influence event.
- Changes after preregistration are allowed only through a dated deviation record stating reason and likely impact.

8. Laboratory operating units

- Visual Stimulus Metrology Unit: captures animation output across device models, brightness levels, refresh rates, color modes, failed unlocks, and accessibility settings.
- Epidemiology and Population Unit: curates epilepsy, photosensitivity, disability-setting, and demographic inputs without treating disability as synonymous with photosensitivity.
- Exposure Modeling Unit: estimates unlocks, eligible animation events, bystander visibility, repeated encounters, and unique-person exposure.
- Adverse Event and Signal Unit: receives complaints, clinical documentation, near misses, and non-seizure effects; separates signal detection from causation.
- Causal Inference Unit: evaluates temporality, stimulus capability, alternative explanations, dechallenge or avoidance evidence, recurrence, clustering, and background rates.
- Institutional Conduct Unit: models denial, complaint fragmentation, non-investigation, procedural delay, nominal compliance, legal resistance, and post-judgment nonresponse.
- Reproducibility and Audit Unit: verifies calculations, seeds, code, evidence labels, and report consistency.

9. Evidence classification and data dictionary

Every numeric field and narrative claim must carry one evidence label: Observed, Published Estimate, Derived Calculation, Projection, Hypothetical Assumption, or Simulation Result. A seventh status, Unknown, is required when the evidence is insufficient.

Each record also stores source, source date, population definition, unit, uncertainty, geography, age scope, collection method, transformation history, version, and analyst. Derived values point to their parent inputs. Simulated values point to scenario ID, seed, and model version.

10. Research questions

- What is the prevalence of active epilepsy and the narrower prevalence of photosensitive or visually provoked seizures?
- What visual properties does the fictional animation exhibit across device and environmental conditions?
- How often are owners and bystanders exposed, and how clustered are repeated exposures?
- What event rates are compatible with observed complaints after accounting for underreporting and background events?
- What sample sizes and exposure counts are required to observe rare outcomes?
- How do general-population and disability-service networks differ without stereotyping disability?
- How much cumulative exposure and modeled harm occurs after each notice milestone when mitigation remains zero?
- What would the counterfactual burden be under opt-out, safe default, redesign, or removal, even though the actual scenario never adopts them?

11. Study design and work packages

Work Package 1 performs literature and standards review. Work Package 2 defines the animation as measurable frame sequences rather than using the vague word flash. Work Package 3 captures device output without presenting it to people. Work Package 4 builds general and enriched synthetic populations. Work Package 5 models primary, household,

occupational, and environmental exposure. Work Package 6 models adverse outcomes and reporting attrition. Work Package 7 evaluates causal evidence. Work Package 8 runs persistent-nonresponse and counterfactual branches. Work Package 9 performs adversarial review and independent replication.

The web-accessibility threshold is a required screen but not the whole medical analysis. W3C Success Criterion 2.3.1 states that content must not flash more than three times in one second unless it is below general and red-flash thresholds. W3C also identifies migraines, dizziness, nausea, and seizures as concerns. The laboratory records compliance by criterion and measured condition rather than using a binary safe label.

Recent research notes that international photosensitivity guidance considers flash rate together with affected area and intensity, and that automated analysis tools themselves require better validation. The laboratory therefore uses multiple analyzers where possible and retains frame-level evidence for manual and independent review.

12. Safe stimulus characterization protocol

- Capture the animation electronically or with shielded instrumentation; do not ask a susceptible observer to watch it.
- Record luminance transitions, saturated-red transitions, spatial area, contrast, duration, frequency, pattern repetition, frame timing, refresh-rate interaction, and display peak brightness.
- Test successful unlock, failed unlock, rapid retries, wake-to-lock transitions, biometric and credential pathways, reduced-motion settings, battery saver, dark mode, and OEM variations.
- Analyze owner-distance and bystander-distance geometries computationally.
- Retain original captures, hashes, device identifiers, OS build, settings, calibration record, and analyzer version.
- Classify each condition as below threshold, above threshold, indeterminate, or not assessed. Never translate below threshold into medically harmless without qualification.

13. Population and exposure framework

The default epidemiological anchor is active epilepsy, not lifetime seizure occurrence. WHO reports around 50 million people worldwide with epilepsy, while the Global Burden of Disease 2021 analysis estimated 51.7 million with a 95% uncertainty interval of 44.9 to 58.9 million and approximately 0.7% active epilepsy in 2021.

Photosensitive epilepsy is modeled separately. The Epilepsy Foundation states that about 3% of people with epilepsy may have seizures triggered by certain flashing lights or visual patterns, with risk influenced by frequency, brightness, contrast, distance, wavelength, and eye state. This is a susceptibility input, not a per-unlock event probability.

The exposure distribution must separate unlocks, pickups, checks, sessions, screen activations, and animation-eligible unlocks. Light, typical, heavy, and very heavy categories are scenario strata, not clinical categories. Bystander exposure is modeled through visibility probability, distance, field of view, and repeated co-presence.

14. Representative and enriched populations

A representative sample estimates population burden. It does not answer an individual worker's local encounter probability when the people encountered are selected by occupation, service setting, household, or care needs.

The laboratory runs at least three population models: a general population; an empirically characterized local or service population using anonymized aggregate data; and sensitivity-only enrichment models such as two, three, or five times a baseline. Enrichment factors are assumptions unless supported by data.

The model must not infer photosensitivity from intellectual, developmental, physical, or psychiatric disability. It may separately model documented epilepsy, known photosensitivity, migraine, vestibular sensitivity, communication barriers, fall consequences, and inability to change settings.

15. Adverse-event and outcome taxonomy

- No observable response.
- Visual discomfort, headache, migraine, nausea, dizziness, disorientation, or sensory distress.
- Aura-like or neurological symptoms without confirmed seizure.
- Suspected seizure, clinically assessed seizure, or seizure with strong time-lock evidence.
- Secondary injury, emergency evaluation, hospitalization, loss of independence, device avoidance, missed services, caregiver burden, or occupational impact.
- Complaint, near miss, unresolved report, legal claim, or regulatory referral.
- Unknown outcome because follow-up or documentation is unavailable.

16. Complaints, adverse events, and evidence preservation

Any person may submit a complaint without having to prove causation. Intake separates what happened, what was observed, what was inferred, and what documentation exists. Reports are never closed merely because the event is rare, the reporter has a preexisting condition, or the laboratory cannot reproduce it immediately.

Triage addresses immediate safety and referral needs without offering medical diagnosis. Duplicate reports are linked rather than erased. False attribution is possible and must be assessed, but it is not presumed. The laboratory publishes aggregate complaint flow, attrition, classification changes, and unresolved counts.

The complaint channel is protected from sponsor, developer, and analyst retaliation. Reporters may correct the record. Data minimization, consent, confidentiality, and access controls apply.

17. Causal inference framework

Causal conclusions are graded, not binary. The laboratory evaluates whether the stimulus is physically capable of producing the proposed effect under the measured condition; whether an event is time-related; whether the person is susceptible; whether competing explanations exist; whether exposure avoidance changes recurrence; whether events cluster after rollout; and whether results replicate.

Possible grades are unsupported, indeterminate, suggestive, probable within a defined subgroup and condition, or strongly supported. A court outcome, settlement, complaint volume, or threshold failure is evidence but does not substitute for the complete causal assessment.

18. Simulation model

The simulation uses person-level or agent-level records with age band, device use, daily unlock distribution, animation eligibility, owner and bystander exposure, epilepsy status, photosensitivity status, other visual sensitivity, individual threshold, susceptibility modifiers, event occurrence, severity, reporting, complaint classification, and device abandonment.

The institutional branch contains evidence state, public position, complaint-handling state, legal state, and mitigation effectiveness. In the requested primary scenario, mitigation effectiveness $M(t)=0$ for every period. The organization may change rhetoric or legal strategy, but exposure is not reduced by voluntary corrective action.

Counterfactual branches set $M(t)$ above zero for an opt-out, safe default, redesign, or removal. They exist only to estimate preventable exposure and harm, not to imply that the fictional organization chooses them.

19. Core equations

- Animation exposures: $E(t) = \text{Users}(t) \times \text{eligible unlocks per user}(t) \times \text{animation-display probability}(t) \times \text{viewer multiplier}(t)$.
- Residual exposure: $E_{\text{residual}}(t) = E(t) \times [1 - M(t)]$. In the primary persistent-refusal branch, $M(t) = 0$.
- At least one event under constant independent per-exposure risk p : $1 - (1 - p)^n$. The model must test dependence and frailty alternatives rather than assume independence.
- Expected events for a simple Poisson approximation: $\lambda = \text{sum of susceptible exposures} \times \text{event probability}$. Report probability of zero as $\exp(-\lambda)$ only when assumptions are appropriate.
- Encounter probability for n independent unique people with prevalence q : $1 - (1 - q)^n$. Use clustered or hierarchical alternatives for households and service settings.
- Preventable events: events in persistent-refusal branch minus events in a specified counterfactual branch.

20. Persistent institutional nonresponse protocol

Persistent refusal is implemented as a deterministic policy constraint, not as a probability that eventually decays. The fictional organization does not voluntarily implement material mitigation at any evidence threshold.

The model records denial modes separately: complete denial, causal deflection, responsibility deflection, statistical minimization, technical minimization, accommodation refusal, complaint fragmentation, legal resistance, symbolic compliance, and post-judgment resistance.

Legal or regulatory milestones do not automatically terminate exposure. External intervention may be modeled in separate branches, such as an injunction, OEM override, operating-system fork, regulator-ordered disablement, insurer pressure, or market exit. These are external terminations and must not be mislabeled as voluntary developer correction.

21. Statistical analysis plan

- Preregister primary outcomes, attribution rules, exclusion rules, model families, sample sizes, stopping rules, and sensitivity ranges.
- Use rare-event binomial, Poisson, survival, hierarchical, Bayesian, or case-crossover methods according to the data-generating process.
- Estimate mean, median, uncertainty intervals, probability of zero, probability of at least one, tail probabilities, and person-level recurrence.
- Run sensitivity to prevalence, unlock distribution, animation eligibility, bystander exposure, reporting probability, susceptibility concentration, dependence, and device abandonment.
- Perform negative-control and background-rate analyses where data permit.
- Never interpret expected counts as guaranteed observations.
- Publish null, indeterminate, and subgroup-limited findings.

22. Stop, pause, and escalation rules

The laboratory does not expose participants to the animation, so its principal stop rules concern data collection, device testing, privacy, and reporting. Any unexpected human exposure, suspected serious adverse event during laboratory activity, privacy breach, instrument malfunction that invalidates results, or undisclosed conflict triggers immediate suspension of the affected workstream.

Because the simulated developer never acts, an independent laboratory safety escalation is not contingent on developer acknowledgment. The lab may issue a public safety notice, notify an

ethics body or regulator, preserve evidence, and recommend avoidance or technical controls, while clearly stating the evidence grade and uncertainty.

23. Audit, reproducibility, and transparency

- Version-control the protocol, code, parameter dictionary, synthetic data generator, and reports.
- Hash raw captures and lock the analysis dataset before unblinding primary results.
- Keep an assumptions register and a deviation register.
- Require independent rerun of all headline results.
- Publish enough detail for external reconstruction without exposing personal data.
- Archive superseded versions and do not overwrite adopted records.
- Label proposed runs separately from executed runs.

24. Review of individual AI contributions

ChatGPT provided the most comprehensive decomposition and correctly required a model capable of finding no hazard, an undetectable hazard, a subgroup effect, or a material signal. Its follow-up supplied the strongest two-system distinction between evidence accumulation and organizational action.

Claude provided the clearest concise lifecycle of refusal and the useful deterministic formulation that the probability of acknowledgment or remediation is zero. Its first rewrite was too brief for the requested scope and assumed elevated prevalence in the user's work population without first requiring evidence.

Perplexity contributed practical modeling elements, underreporting, safe simulation, and an explicit mitigation parameter. Its early answer still framed nonresponse as an ethical comparison against recommended action, rather than the permanent primary branch. The follow-up corrected this well.

Grok contributed useful stimulus skepticism, a zero-risk arm, event-attribution criteria, and the observation that an instrument unable to detect seizures can manufacture an absence of signal. Some factual and numerical claims in the supplied response are uncited in the compilation and must be independently verified before use.

Microsoft Copilot provided strong evidence-labeling, primary and bystander exposure, enriched-network analysis, and a taxonomy of denial modes. Its initial wording used "initially declines," which understated the requested persistent condition; the follow-up corrected that directly.

No single response is adopted wholesale. The charter combines their strongest structural contributions and rejects unsupported precision, stereotypes, real-company allegations, and any assumption that evidence automatically produces organizational correction.

25. Deliverables and completion gate

- Preregistered protocol and analysis plan.
- Stimulus metrology report with frame-level evidence.
- Evidence and assumptions register.
- General-population and enriched-population exposure models.
- Five-year persistent-refusal simulation and counterfactual comparison.
- Complaint and adverse-event aggregate report.
- Causal-inference assessment.
- Reproducibility package and independent rerun report.
- Public executive summary written for mobile reading.
- Completion requires clear purpose, labels, preparer, date and timezone, non-claims, evidence labels, numbered exhibits, consistent Markdown/DOCX/PDF substance, and no claim that an unexecuted simulation was run.

26. Adoption and change control

This document is a draft. It may guide the next design step but is not adopted, accredited, validated, or canon. Adoption requires a named coordinator or governing body. Material changes require a new version, revision note, and preservation of the prior version. The laboratory may not self-certify independence or accreditation merely by following this charter.

27. Sources and related documents

- Primary source compilation: ANDROID UNLOCK, ANIMATION, SEIZURE RISK, ACCESSIBILITY SIMULATION.pdf.
- Documentation standard: OAL-ONE-LC_Documentation-Standard_Draft_v0.1_2026-10-04.pdf.
- WHO epilepsy fact sheet: <https://www.who.int/news-room/fact-sheets/detail/epilepsy>
- GBD 2021 epilepsy analysis: <https://www.thelancet.com/journals/lanpub/article/PIIS2468-2667%2824%2900302-5/fulltext>
- W3C Understanding WCAG 2.3.1: <https://www.w3.org/WAI/WCAG21/Understanding/three-flashes-or-below-threshold.html>
- Epilepsy Foundation photosensitivity guidance: <https://www.epilepsy.com/what-is-epilepsy/seizure-triggers/photosensitivity>
- Jordan and Vanderheiden guidelines gap analysis: <https://dl.acm.org/doi/epdf/10.1145/3694790>
- Evaluating conformance of video safety tools: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12249941/>
- ISO/IEC 17025 impartiality overview used as orientation, not a claim of accreditation: <https://advisera.com/17025academy/blog/2020/10/12/ensuring-impartiality-in-an-iso-17025-laboratory/>

28. Revision note

v0.1, 2026-10-04. Initial independent draft. No simulation executed. Not reviewed or adopted.